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{European Digital Innovation Hubs – Analysis and Development of the
Sectors, Services, Technologies Framework}

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Abstract	4
Introduction.....	6
Research Objectives and Contributions	9
1 EDIH Ecosystem	11
1.1 Evolution and Core Functions of European Digital Innovation Hubs (EDIHs)	11
1.1.1 Enhanced Benefits of the EDIH Ecosystem	11
1.2 Challenges and Disparities in the EDIH Landscape	15
1.2.1 Operational and Strategic challenges	15
1.2.2 Limitations of Previous Research and the Need for a Novel Approach	17
1.3 Existing Analytical Frameworks for EDIH Assessment	18
1.3.1 The D-BEST References Model and its Evolution	18
1.3.2 DR-BEST (Data-Remote-Business-Ecosystem-Skills-Technology)	19
1.3.3 L-BEST (Legal and Ethical-Business-Ecosystem-Skills-Technology)	21
1.3.4 Alternative Frameworks for EDIH Assessment	22
1.4 Cluster Analysis Concepts, Methods, and Applications in EDIH Research.....	22
1.4.1 Clustering Concepts and Methods	22
1.4.2 Comparison of Clustering Methods for EDIH Data	28
1.4.3 Use of Clustering in EDIH Research and Policy	30
1.4.4 Tools and Libraries for Clustering EDIH Data.....	32
1.4.5 Effectiveness of Clustering Methods for EDIH Analysis.....	34
2 Methodology.....	37
2.1 Data Collection	38
2.2 Data Preprocessing.....	39
2.3 Analytical Methods.....	43
2.4 Framework Development.....	45
3 Results	47
4 Discussion.....	47
5 Conclusion	48
5.1.1 Research Objectives and Contributions	48
5.1.2 Methodology	48

6	References	49
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Abstract

One of the key components of the European Union's digital transformation policy under the European Digital Innovation Hubs (DIH) is to enhance digital technology adoption in various sectors of existing and emerging industries. These hubs give support to improve businesses products as well as services using digital technologies with the help of granting access to technical expertise, testing, innovations services and sustainability and circularity.

Despite the great initiative, gaps exist in meeting sectoral demands, implementation of innovative technology, addressing respective economic of issues. In this study the EDIH ecosystem will be analyzed by mapping sectors, services, and technological capabilities. Thirty-eight (38) economies will be analyzed (EDIH catalogue from the European Commission) from manufacturing, healthcare, agriculture, energy, and logistics using clustering methods and desk research. Ultimately clusterization of EDIH ecosystem will be proposed accompanied with the framework of sector-service-technology (SST) matrix implying policy recommendations to address regional and sectoral gaps. It may serve to assess alignment with regional, national, and European policies, and technological trends. This study will build up on previous research made by highlighting sector-specific activities, trends in technological take and influence of the regional and EU policies on their efficiency. Additionally, the research will also focus the analysis of EDIH based on Circular economy. EDIHs supposed to serve as a foundation for the transition towards more sustainable practices by the implementation of novel technologies, support knowledge exchange and by helping businesses adapt to the evolving economic conditions.

The study applies cluster analysis to classify DIHs based on their sectoral and technological offerings and regression modeling to examine the factors influencing service diversity. Qualitative insights are drawn from case studies across different European regions, providing deeper perspectives on operational challenges, funding mechanisms, and policy-driven digitalization efforts. The study tries to compare and analyze the service and technological offerings of the DIHs with the regional economic outlook and deficiencies

Sector-Specific DIH Activities

This research intends to contribute by making a sectoral analysis of DIHs, assessing how they align with industry-specific digital transformation needs. A European DIHs exhibit considerable

variation in their focus areas, often reflecting the economic strengths of their respective regions. Studies indicate that industrially advanced economies tend to prioritize Industry 4.0 solutions, such as smart manufacturing and automation, while rural regions focus on AgriTech and digital solutions for sustainable agriculture (Gyulai et al., 2024). However, emerging sectors like green technology remain underrepresented, even as sustainability initiatives gain traction.

Research by Gavkalova & Gładysz(Gavkalova et al., 2024) supports this observation, demonstrating that DIHs in Western and Northern Europe(France, Germany, Italy, the Netherlands, and Spain) are more diversified in their service portfolios, while Eastern and Southern European(Greece, Lithuania) hubs often specialize in more narrow, sector-specific offerings. Their findings emphasize that economies with a higher number of DIHs tend to exhibit more balanced service provision, whereas smaller economies rely heavily on centralized hubs, which can lead to accessibility and scalability challenges.

Another critical insight comes from Sassanelli(Asplund et al., 2021a), who problematize the service portfolio of DIHs, arguing that many hubs struggle with aligning their services with regional industry demands. Their study suggests that the misalignment of DIH services and industry needs can hinder digital adoption, particularly in SMEs that require tailored solutions.

Technology Adoption Patterns Across DIHs

The technological landscape of DIHs varies significantly across Europe, with some regions demonstrating a higher propensity for AI, IoT, and blockchain integration, while others focus on cybersecurity and cloud services. Research by Thelen et al. (2024) indicates that hubs in high-GDP economies tend to prioritize cutting-edge AI applications, particularly in smart manufacturing and digital finance. Meanwhile, DIHs in less economically developed regions often focus on foundational digital infrastructure, reflecting funding limitations and skill shortages.

Introduction

Innovation in the modern global landscape is more than simple technological progress; it represents a collective response to profound challenges such as fierce international competition, pressing climate issues, and entrenched inequalities. It embodies a commitment to building strong, connected communities where people, ideas, and resources come together, often leveraging new technological paradigms and data-driven insights, to effectively solve real-world problems. This broader understanding of innovation, driven by the affordances of emerging technologies, is critical for addressing complex societal missions (Nambisan et al., 2017).

In these rapidly changing times, innovation represents more than having the latest devices or systems; it's about building strong, interconnected societies where people, ideas, and resources come together to solve real problems (Horgan & Dimitrijević, 2018). The innovation ecosystem is central to overcoming these challenges - a collaborative network where diverse stakeholders, businesses, academic institutions, and government bodies interact to stimulate technological and economic growth. These ecosystems serve as pivotal centers for knowledge exchange, resource sharing, and the cultivation of digital transformation. The effective management of these intricate ecosystems is paramount to ensuring their long-term sustainability and impactful contributions (Thomas & Autio, 2019). This management involves bridging existing gaps between stakeholders, aligning regional capabilities with global challenges, and guaranteeing equitable access to essential infrastructure and expertise, particularly for regions that have historically been underserved (Granstrand & Holgersson, 2020).

For the European Union (EU), this mission carries particular significance as it endeavors to maintain its economic leadership while simultaneously navigating a substantial green and digital transformation. Bold initiatives such as the Digital Decade aim for a substantial adoption rate, targeting 75% of EU enterprises to utilize cloud and Artificial Intelligence (AI) solutions by 2030. (*Europe's Digital Decade: 2030 Targets* | *European Commission*, n.d.) However, the reality reveals a fragmented innovation landscape, where numerous regions contend with uneven access to specialized expertise, cutting-edge infrastructure, and the requisite funding, collectively impeding progress toward these ambitious digital goals. (*How Digitalised Have the EU's Enterprises Become? - News Articles - Eurostat*, n.d.)

Digital innovation Hubs (DIHs) emerged as a direct policy response to these challenges, initially conceived under the EU's Digitizing European Industry (DEI) initiative in 2016. These hubs

were specifically designed to bridge the chasm between research, technology, and industry with a particular focus on supporting Small and Medium-sized Enterprises (SMEs). They provided localized access to digital tools, specialized expertise, and crucial funding intended to democratize digital transformation across the European continent *Make the Most of Digital Opportunities — Brochure*, n.d.).

Despite their initial pledge, standalone DIHs encountered significant limitations, including geographic fragmentation, inconsistencies in service quality, and a noticeable lack of cross-border collaboration. To overcome these inherent limitations, the EU launched European Digital Innovation Hubs (EDIHs) under the Digital Europe Program in 2021 (Digital Innovation Hubs as Pol(Volpe et al., 2021).

EDIHs represent a significant evolution from their predecessors, structured as a multi-tiered, pan-European network designed to integrate regional hubs into a cohesive and interconnected ecosystem (Serrano-Ruiz et al., 2024). This evolution from localized DIHs to a comprehensive, networked EDIH ecosystem reflects a deeper understanding of systemic innovation challenges. It acknowledges that isolated local efforts, while valuable, are insufficient to drive continent-wide digital transformation and address pervasive disparities. The strategic shift from individual "hubs" to an integrated "network of hubs" signifies a move from localized problem-solving to a more sophisticated, systemic approach to ecosystem management, aiming to foster synergistic effects and leverage collective intelligence across national borders (Serrano-Ruiz et al., 2024).

Unlike earlier DIHs, EDIHs operate across four distinct hierarchical levels—regional, national, transnational, and EU-wide—to ensure robust alignment with broader EU sustainability goals and critical sectoral priorities. This structural transformation underscores the imperative to manage innovation ecosystems at scale, effectively balancing localized needs with overarching strategic objectives such as the Green Deal and the Digital Single Market. For a DIH to achieve official recognition, it must adhere to specific criteria, including integration into a regional, national, or European policy initiative focused on industrial digitization. Furthermore, it must function as a non-profit organization, maintain a physical presence within its designated region, operate an updated website, and provide publicly accessible case studies demonstrating its support for at least three companies in their digitization endeavors *icy Instruments to Boost Digitalisation of SMEs - Publications Office of the EU*, n.d.).

European Digital Innovation Hubs serve as the foundational backbone of the EU's overarching digital transformation strategy, providing SMEs with indispensable access to technical expertise,

advanced technologies, and crucial testing facilities. The strategic importance of SMEs cannot be overstated; they constitute 99% of all EU businesses, collectively employing over 85 million citizens and generating an impressive €4 trillion in value themselves. Crupi et al.,

2020)(*Innovation and Digital Transformation: How Do European SMEs Perform?*, n.d.)

Consequently, their successful digitization is not merely beneficial but critically important for ensuring Europe's economic resilience and competitiveness on a global scale.

Despite the substantial efforts invested in these initiatives, only 58% of SMEs had achieved basic digital intensity by 2023, a figure significantly below the EU's ambitious 2030 target of 88%

(*Digitalisation in Europe – 2024 Edition - Interactive Publications - Eurostat*, n.d.). This

persistent gap highlights a critical disconnect between the supply-side support offered by EDIHs and the demand-side absorption capacity of SMEs. The observed disparity suggests that while EDIHs provide a robust supply of digital assistance, SMEs may face fundamental barriers—such as a lack of awareness, internal skill shortages, financial constraints beyond direct EDIH support, or cultural inertia—that prevent them from fully leveraging these services. This situation underscores the urgent necessity for optimizing EDIHs to more effectively address sector-specific demands, mitigate regional disparities, and overcome existing policy barriers. A more granular analysis of SME needs and adoption hurdles, beyond merely the availability of services, is clearly indicated.

Academic and policy research has been examining the role and impact of European digital innovation hubs in Europe's digital ecosystem. (Asplund et al., 2021b) observed that networking opportunities provided by hubs help start-ups form strategic partnerships and access new markets. Similarly, other recent works (Sassanelli & Terzi, 2022a) (Quadrini et al., 2022) that an EDIH can offer a **broad spectrum of assets and services** – from technology infrastructure to business mentorship – acting as a “knowledge broker” in its region. To systematically describe these offerings, researchers have proposed frameworks like the **D-BEST reference model**, which categorizes a hub's services into five macro-classes: **Data, Business, Ecosystem, Skills,** and **Technology**. The D-BEST model provides a structured lens to compare hubs' service portfolios and has been used to analyze how DIHs support Industry 4.0 collaborations. Overall, the existing literature underlines that EDIHs are expected to **bolster innovation capacity** of SMEs and drive Europe toward digital leadership in the Industry 4.0 era.(Sassanelli & Terzi, 2022b)

Despite growing interest, significant knowledge gaps remain regarding the pan-European landscape of EDIHs and their effectiveness. First, while individual case studies and regional analyses of digital innovation hubs exist, we lack a **comprehensive mapping of all EDIHs across Europe and a comparative assessment of their service offerings**. As recent scholars point out, literature still does not provide a full panorama of how these hubs are distributed, what specializations they have, and how they differ in various economic contexts. For instance, the D-BEST framework mentioned above has so far been applied only in limited scopes – e.g. to compare a couple of countries or specific hub networks – without investigating broader patterns or the causes behind different hub configurations. Moreover, the research examined two EU hub networks (in the context of EU innovation programs) and identified the key service functions, but their study was bounded to those cases and did not chart the diversity of hubs EU-wide (Asplund et al., 2021c).

Likewise, (Gavkalova et al., 2024) conducted an initial cross-country analysis of DIH service portfolios, clustering European economies by the types of services their hubs offer. Their findings revealed that hubs in different countries emphasize different services (reflecting varying national digitalization strategies. Yet even this study acknowledges the need for “*more precise indications about [the] status and nature*” of DIHs across all economies.

In particular, existing research has **not addressed the interplay between the sectoral focus of hubs, the specific services they provide, and the technologies they specialize in**. Most studies consider *what* services hubs offer, but not *in which sectors* these services are applied or *which technologies* (e.g. AI, IoT, cybersecurity, etc.) are emphasized. This multidimensional view – linking sector, service, and technology – remains underexplored.

However, the impact of contextual factors—such as a region’s economic profile or level of digital maturity—on a hub’s service portfolio remains poorly understood, and the literature still lacks a truly integrative, multi-dimensional analysis of the European EDIH network that clearly describes what hubs do in terms of service offerings, examines how they leverage specific technologies, and explores why certain patterns emerge in relation to contextual drivers. This thesis therefore provides a comprehensive investigation of service–technology–context interactions within the European EDIH ecosystem, explicitly narrowing its scope by foregoing any country-by-country mapping of hubs to national digitalization strategies.

Research Objectives and Contributions

This research aims to uncover analytical insights into European Digital Innovation Hubs (EDIHs) by analyzing their service offerings using a Sector–Service–Technology (SST) matrix. The SST matrix provides a novel visualization of each EDIH’s offerings across sectors, services, and technology domains, highlighting patterns within the EDIH ecosystem. Building on these insights, the study identifies gaps in the current EDIH landscape and proposes policy recommendations to address them. Furthermore, the analysis focuses on selected geographical regions that belong to the same cluster grouping, ensuring comparability across similar contexts. To achieve these aims, the research is structured around three detailed objectives:

- **Comprehensive EDIH Mapping:** Compile and analyze a **pan-European database of EDIHs**, detailing each hub’s sectoral specializations, service offerings, and key technology domains.
- **Development of Sector–Service–Technology (SST) Matrix** that provides a novel visualization of how each EDIH (and the network as a whole) aligns industry sectors with services and digital technologies. This SST Matrix offers a new tool for policymakers and scholars to identify coverage gaps and overlaps in the EDIH network.
- **Identification of EDIH Typologies:** Using data-driven analysis, identify **distinct clusters or typologies of EDIHs**. The clustering is expected to reveal groups of hubs with similar profiles (for instance, a cluster of hubs focused on manufacturing-sector SMEs with strong robotics/AI services, versus another cluster focusing on digital skills training in multiple sectors, etc.). Uncovering these patterns will help explain how different regional or sectoral needs are being addressed by hubs. It also provides practical insights: EDIHs within the same cluster could share best practices, and policymakers could tailor support knowing one type of hub differs from another.
- **Policy Implications and Recommendations for Selected Economies:** Synthesize the research findings to extract policy implications and formulate targeted recommendations for selected European economies. This objective translates the study’s insights into actionable guidance for policymakers, addressing identified digital innovation gaps and aligning EDIH initiatives with both national and EU strategic goals.

1 EDIH Ecosystem

1.1 Evolution and Core Functions of European Digital Innovation Hubs (EDIHs)

Digital Innovation Hubs (DIHs) serve a crucial role in accelerating the adoption of advanced technologies, including Artificial Intelligence (AI), Internet of Things (IoT), cybersecurity, big data and blockchain, with a particular emphasis on Small and Medium-sized Enterprises (SMEs) and the public sector. DIHs primarily fulfill four essential functions in supporting the digitization of SMEs: fostering robust networking, providing comprehensive skills and training programs, enabling "test before investing" opportunities, and facilitating access to crucial funding. (Kalpaka A et al., 2022)

Beyond merely facilitating digital transformation, DIHs act as pivotal enablers of innovation by actively fostering collaboration among businesses, research institutions, and technology providers. They play a supportive role in helping companies secure necessary investments, refine their business models, and ultimately enhance their competitiveness through strategic digital advancements. A fundamental function of DIHs is also skills development, which ensures that both businesses and individuals acquire the requisite expertise to effectively leverage emerging technologies. Furthermore, they offer pre-investment testing, granting businesses access to specialized testing facilities and expert guidance to mitigate the inherent risks associated with adopting new digital solutions. (Sassanelli & Terzi, 2022a)

1.1.1 Enhanced Benefits of the EDIH Ecosystem

The evolution from standalone DIHs to European Digital Innovation Hubs (EDIHs), initiated under the Digital Europe Programme, has culminated in the establishment of a multi-level relational network. This network seamlessly integrates regional hubs into a cohesive, EU-wide ecosystem. (Serrano-Ruiz et al., 2024). This formalization of a multi-tiered relational network, from regional DIHs to a pan-European EDIH structure, signifies a strategic move towards institutionalizing cross-border knowledge transfer and optimizing resource allocation. (Volpe et al., 2021) This is critical for addressing fragmented innovation landscapes and achieving synergistic effects across the continent. The hierarchical structure is designed to overcome the

inherent limitations of isolated innovation efforts, recognizing that complex challenges like AI adoption or sustainability require coordinated, multi-scalar responses. (Serrano-Ruiz et al., 2024)

This comprehensive network operates across four distinct hierarchical levels, each contributing uniquely to the overarching European innovation landscape:

- **DIH Level:** At this foundational tier, individual Digital Innovation Hubs (DIHs) function as centralized hubs within their respective regional frameworks. Here, organizations collaborate to enhance their competitiveness through the strategic application of digital technologies. These hubs integrate a diverse array of stakeholders—including universities, research centers, private technology providers, regional development agencies, and governmental bodies—and deliver tailored support services, such as process optimization, product innovation, and direct access to digital tools. ((Serrano-Ruiz et al., 2024)
- **DIH Ecosystem:** Each DIH serves as the anchor for a regional innovation ecosystem, effectively connecting technology providers with organizations actively seeking digital transformation. While SMEs are the primary beneficiaries, larger entities requiring specialized solutions also engage within this ecosystem. This level places significant emphasis on knowledge exchange, successfully bridging the gap between cutting-edge technological advancements and their practical applications within various industries. (Serrano-Ruiz et al., 2024)
- **EDIH Level:** Building upon the foundation of regional DIHs, European Digital Innovation Hubs (EDIHs) aggregate multiple hubs into transnational collaborative networks. These networks transcend national borders, enabling robust cross-country partnerships to address shared challenges, including sustainability, AI adoption, and the development of crucial digital skills. By pooling collective resources and expertise, EDIHs significantly amplify the impact of individual DIHs while ensuring alignment with broader EU strategic priorities.(Serrano-Ruiz et al., 2024)
- **Pan-European EDIH Network:** This represents the highest tier of the network, a continent-wide cooperative framework where various EDIH groups—even those that might otherwise be in competition—establish formal channels for collaboration. This "coopetition" model strategically balances competitive funding dynamics with collective advocacy for shared interests, such as influencing policy or engaging in broader societal initiatives. The presence of "coopetition" at the highest tier indicates a mature stage of

ecosystem development where even competing entities are compelled to collaborate on shared interests, fostering a robust and resilient system for innovation diffusion and capacity building across diverse European regions.(Serrano-Ruiz et al., 2024)

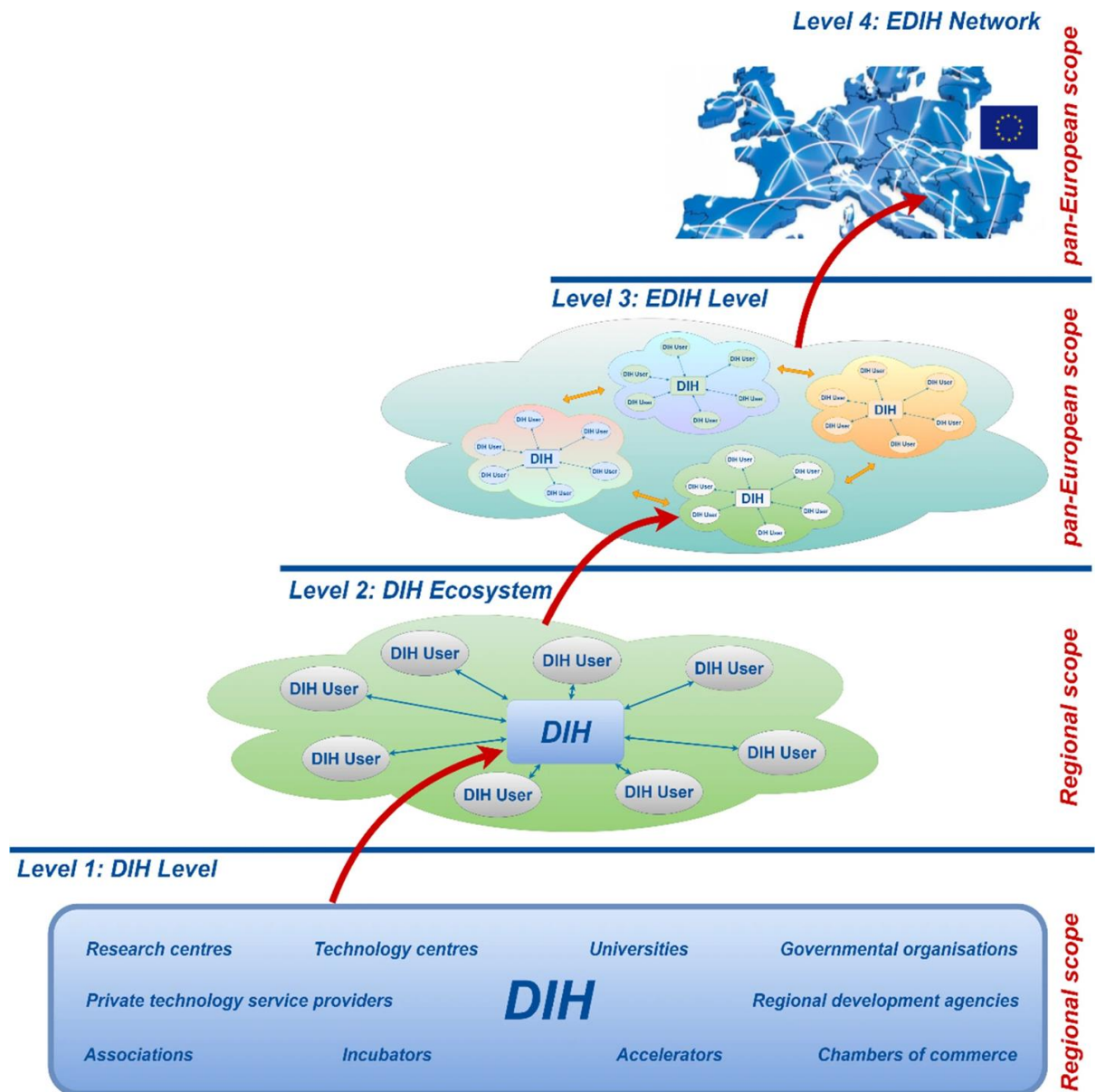


Figure: relationship levels within the pan-European space of DIHs and EDIHs action (Serrano-Ruiz et al., 2024)

The newly established EDIH ecosystem delivers four clearly defined benefits, demonstrating a shift from purely technological support to a more holistic, risk-mitigating, and commercially oriented approach for SMEs:

- **Testing before Investing:** SMEs gain crucial access to state-of-the-art testing and experimental facilities for their technological developments or for validating the proof of concept for new business ideas, allowing them to assess feasibility before committing significant expenditure. This capability facilitates informed decision-making for future investments and enables experimentation with digital advancements within dedicated testbeds, sandboxes, and pilot programs. This explicit emphasis on de-risking adoption acknowledges the practical economic hurdles SMEs face. (*European Digital Innovation Hubs in Digital Europe Programme*, 2019)
- **Skills and Training:** EDIHs actively support businesses by fostering expertise through continuous learning processes and specialized training programs tailored for their workforce. These programs are meticulously designed based on specific sectoral needs, with a particular focus on new and relevant technologies and their practical implementation plans. (*European Digital Innovation Hubs in Digital Europe Programme*, 2019)
- **Support to find Investment:** Through the extensive EDIH network, SMEs are empowered to explore new funding opportunities and receive expert advice on how to craft successful business plans and compelling investment pitches for potential investors. They also gain access to a wealth of resources, including comprehensive guides and best practices, thereby enhancing their understanding of complex financial issues. By integrating pre-investment testing and financial advisory, EDIHs move beyond being mere technology providers to becoming comprehensive business accelerators, directly addressing the financial and strategic uncertainties that often impede digital transformation for smaller enterprises. (*European Digital Innovation Hubs in Digital Europe Programme*, 2019)
- **Innovation Ecosystem and Networking:** EDIHs actively promote increased collaboration with diverse stakeholders and other EDIHs across regional, national, and European networks, with the overarching goal of facilitating extensive knowledge exchange. They organize a variety of workshops and events designed to inspire and motivate participants, utilizing matchmaking services to connect them with other private and public actors that align with their specific needs. (*European Digital Innovation Hubs in Digital Europe Programme*, 2019)

Overall, EDIHs function as "end-to-end" centers, providing a comprehensive suite of services that encompasses skills development, financial advisory support, and pilot testing for circular economy

solutions. When regional EDIHs identify a lack of specific expertise within their immediate network, they strategically leverage their broader networked structure to connect SMEs with highly specialized hubs located across Europe.

1.2 Challenges and Disparities in the EDIH Landscape

1.2.1 Operational and Strategic challenges

Despite their significant potential to drive transformation, European Digital Innovation Hubs (EDIHs) encounter a range of operational and strategic challenges that constrain their full effectiveness. These impediments primarily stem from several interconnected factors which are discussed in the following:

- **Regional disparities in digital infrastructure:** A notable unevenness exists in the distribution and quality of digital infrastructure across various European regions. This directly impacts the capacity of EDIHs to deliver advanced services uniformly and efficiently (Capello & Caragliu, 2024).
- **Uneven service portfolios:** The range and depth of services offered by EDIHs are not consistently comprehensive or uniformly distributed across the entire network. This often leads to significant misalignments with the specific digital transformation needs of different regions and industries. (Gavkalova et al., 2024)
- **Lack of sector-specific alignment:** EDIHs may inadvertently overlook or inadequately address the needs of certain sectors by disproportionately prioritizing others. For example, an EDIH located in a region dominated by manufacturing might heavily emphasize industrial automation technologies, potentially neglecting the distinct requirements of Agri-Tech SMEs that need specialized precision farming tools (Gavkalova et al., 2024).
- **Connectivity and technical capacity:** Underdeveloped regions, particularly those in Central and Eastern Europe, frequently lack the necessary broadband connectivity and the technical capacity among their local businesses and workforce to fully utilize the advanced services offered by EDIHs. This exacerbates existing geographic inequalities in digital adoption. (Obelovska et al., 2025)
- **Policy fragmentation:** Misalignments between national digital priorities and broader EU-wide sustainability targets often result in funding mismatches and significantly limit the scalability of successful EDIH initiatives across national borders (European Commission

Data Reveals How European Innovation Is Fragmented - European Commission, n.d.). At the national level, Member States frequently channel digital funding into standalone infrastructure or innovation catch-up programmes, while green and cohesion funds are earmarked separately for climate-related investments, leaving hybrid digital-sustainability actions under-resourced (Gao, 2025). These divergent funding streams undermine the ability of EDIHs to pool resources for pan-European testbeds and cross-border pilot projects, because differing eligibility rules and grant priorities create administrative hurdles and disincentivize joint applications (Publications Office of the EU, 2025). Moreover, the compartmentalization of funding within siloed ministries—ICT versus environment—forces EDIHs to navigate multiple, often conflicting, application frameworks to secure complementary grants, delaying deployment timelines and eroding potential economies of scale (Gao, 2024; Publications Office of the EU, 2025).

- **Misalignment of services with regional industry demands:** Many hubs struggle to effectively align their service offerings with the specific demands of regional industries. This fundamental misalignment can substantially hinder digital adoption, particularly for SMEs that require highly tailored solutions to their unique operational contexts. The recurring theme of "misalignment"—whether it pertains to sector-specific needs, regional demands, or policy objectives—suggests a fundamental issue of contextual relevance and adaptability within the current EDIH framework. This indicates that a "one-size-fits-all" approach to digital transformation support is inherently ineffective. The success of EDIHs critically depends on their ability to perform a sophisticated matching function, aligning available technologies and services with the genuine, context-specific demands of SMEs and the unique structures of regional economies (Gavkalova et al., 2024)(Asplund et al., 2021a).
- **Funding limitations and skill shortages:** DIHs in less economically developed regions often focus on foundational digital infrastructure, which reflects inherent constraints in available funding and a pervasive lack of skilled personnel necessary for advanced digital adoption. (Casorati & Verbeek, 2022)
- **SMEs' capacity to adopt tools:** There are persistent discrepancies between the sophistication of EDIHs' technological infrastructure, such as high-performance computing testbeds, and the actual capacity of SMEs to effectively adopt and integrate these tools.

This often stems from internal skill shortages, financial constraints, or policy inertia within the SMEs t

- **Underrepresentation of emerging sectors:** Emerging sectors, such as green technology, remain notably underrepresented in the focus areas of DIHs, even as sustainability initiatives gain increasing traction and strategic importance at the EU level. This indicates a lag in adapting to new strategic priorities and a potential missed opportunity for fostering sustainable innovation (Meijer, 2024)

1.2.2 Limitations of Previous Research and the Need for a Novel Approach

While the foundational analysis provided by (Gavkalova et al., 2024) on Digital Innovation Hubs (DIHs) across 38 European economies offered valuable initial insights, their study exhibited critical limitations that the current research endeavors to address.

Key Limitations Identified:

- **Data Reliance and Potential Biases:** Their reliance on self-reported data from the Smart Specialization Platform (S3) and manual web scraping of DIH websites introduced potential biases. Service portfolios were categorized based on declared offerings rather than empirical validation of their operational impact, raising questions about the accuracy and completeness of the data utilized.
- **Exclusion of EDIHs:** The study's exclusion of European Digital Innovation Hubs (EDIHs) was a significant limitation. This exclusion was primarily due to the concurrent launch of EDIHs during the study period, which severely restricted the applicability of their findings to the EU's rapidly evolving and policy-driven innovation ecosystem. The focus remained predominantly on the older DIH model.
- **Limited Granularity of Analysis:** (Gavkalova et al., 2024) primarily employed hierarchical clustering of economies based on DIH prevalence and service distribution using the D-BEST model. While this approach is valuable for broad comparisons, it did not delve into the granular sector-service-technology interdependencies that are crucial for a comprehensive and actionable assessment of innovation hubs.

These limitations particularly the reliance on potentially biased self-reported data and the exclusion of the more recent EDIH model, underscore the critical importance of robust

methodological rigor and dynamic data collection in assessing rapidly evolving innovation ecosystems. In such environments, static datasets quickly become obsolete. The very nature of “innovation hubs” implies a continuous evolution of their services and Technologies. Therefore, any robust analysis necessitates not only comprehensive data but also methods capable of capturing this dynamism and validating reported information against operation realities. To address these critical challenges and provide a more nuanced and accurate understanding of the contemporary landscape, this study employs a novel analytical tool: the sector-service-technology (SST) matrix.

1.3 Existing Analytical Frameworks for EDIH Assessment

1.3.1 The D-BEST References Model and its Evolution

The Data-based Business-Ecosystem-Skills-Technology (D-BEST) is a reference model to categorize DIHs service portfolio, developed, validated and applied along the years in several innovation action projects funded by the European Commission. It resulted from several iterations of development and originally generated from the model “ETB” which was grounded on three main macro-classes of services (Ecosystem, Technology and Business). The ETB was enriched with the skills and data macro-classes, strictly needed to better answer to the digital needs triggered. Recognizing the additional needs of Industry 4.0, the ETB model was expanded by adding two new dimensions, Skills and Data, to comprehensively cover talent development and data-driven services. Recognizing the additional needs of Industry 4.0, the ETB model was expanded by adding two new dimensions, Skills and Data, to comprehensively cover talent development and data-driven services. (Quadrini et al., 2022)

The D-BEST model is structured as a three-level taxonomy of services. At the top level are the five **macro-classes** of services (Data, Business, Ecosystem, Skills, Technology), which together span the full range of support activities a hub can provide. Each macro-class is further broken down into **types** of services, and each type is detailed into specific **service classes**. (Sassanelli & Terzi, 2022c)

This hierarchical breakdown makes the model flexible yet granular: for example, the *Technology* macro-class can include service types like “test-before-invest” infrastructure access, prototyping support, or technology validation, each of which is subdivided into concrete service classes (e.g. providing lab facilities, developing proof-of-concepts, product certification support).

Likewise, the *Business* macro-class covers services for later-stage needs such as business model innovation, fundraising assistance, and market access, with its own set of types and classes (Sassanelli & Terzi, 2022c).

In total, the standard D-BEST catalog encompasses five macro-classes subdivided into various service types and numerous specific service classes, covering everything from data acquisition and analysis services to ecosystem-building activities.

This rich classification provides a common language for hubs to describe what they offer (Sassanelli & Terzi, 2022c). Notably, *D-BEST* was designed to align closely with the EU's official DIH/EDIH service categories – it essentially maps onto the four core DIH functions (innovation ecosystem networking, skills training, test-before-invest, and access to finance) while adding “Data” to address the rising importance of data-driven services (Sassanelli & Terzi, 2022c).

1.3.2 DR-BEST (Data-Remote-Business-Ecosystem-Skills-Technology)

The D-BEST framework has been continuously extended and refined to capture the evolving complexities and expanding scope of digital innovation services, reflecting the dynamic nature of the digital economy. The DR-BEST variant introduces a sixth macro-class, **Remote** to account for services delivered without physical interaction, alongside the existing five macro-classes: Data, Business Ecosystem, Skills, and Technology. This addition, made during the AI REGIO project (2020–2022), responds to the heightened demand for remote support (accelerated by COVID-19 restrictions) and aims to encompass the offerings of Didactic Factories and virtual hubs (Razzetti et al., 2022b).

(Razzetti et al., 2022b) classify DIH services into six macro-classes:

- **Data** – services related to data acquisition, processing, analytics, decision support, data sharing, and related “physical-human” interactions.
- **Remote** – services enabling experimentation and support delivered off-site (e.g. data spaces, *ICT-as-a-Service* platforms, digital twins, teleoperation or assets offered as a service).
- **Business** – services supporting business development such as incubation, access to finance, project development, and business training.

- **Ecosystem** – services for community building, innovation ecosystem development, and governance of partnerships.
- **Skills** – services for skills assessment and improvement, including training programs and competency development for workforce and students.
- **Technology** – technical services like ideation support, contract research, providing infrastructure or testbeds (“test before invest”), scale-up assistance, and technology validation and verification. (In earlier D-BEST usage, “Technology” implicitly included data-oriented services; in some projects this was referred to as a combined Technology & Data class.



Figure DR-BEST taxonomy (Razzetti et al., 2022b)

Each macro-class is further broken down into second –level service types and third level specific services. In total, the DR-BEST catalogue identifies on the order of 69 district serves at level 3, spanning the six categories. This standardized taxonomy ensures a comprehensive view of all possible services a hub could offer, while using consistent definitions.(Razzetti et al., 2022b)

Regarding the roles DR-BEST provides a reference model to DIHs to analyze and configure their service portfolio in a structured way. By classifying existing services into the DR-BEST categories, a hub can identify gaps in it’s offerings (missing service types) and areas for improvement or expansion. It also allows for benchmarking and alignment across hubs; if every DIH describes its services using the same framework, it becomes easier to compare portfolios and ensure all key support areas(Data, Business, etc) are covered. Moreover the inclusion of the Remote dimension

in DR-BEST is especially relevant as DIHs increasingly offer virtual services and platforms. It highlights the “remotization” aspect of digital support, ensuring hubs consider online platforms, simulators and cloud-based services as part of their offerings. This is crucial for enabling inter-hub service interoperability and reaching companies beyond the hubs’s immediate region(Razzetti et al., 2022b).

1.3.3 L-BEST (Legal and Ethical-Business-Ecosystem-Skills-Technology)

L-BEST is a specialized extension of the D-BEST taxonomy to address emerging legal and ethical considerations in Artificial intelligence services. As AI adoption grew, DIHs specializing in AI needed to support SMEs not only technically and business-wise, but also in navigating AI’s legal requirements (e.g. GDPR, data privacy laws) and ethical issues (bias, transparency, safety) (Razzetti et al., 2022a). L-BEST, the acronym for the resulting framework, denotes the inclusion of Legal-Ethical services in addition to the conventional Business, Ecosystem, Skills, and Technology programs (ceur-ws.org). Notably, the categories of technology and data are combined in L-BEST; the "T" stands for technology and data services, allowing the "L" to be included while maintaining the five-letter acronym ceur-ws.org. In response to emerging AI regulatory requirements, L-BEST was created with the goal of educating DIHs on the legal and ethical competency areas they need to cultivate in order to responsibly promote AI innovation(Razzetti et al., 2022a).

The model consists of five macro-classes of services: Legal & Ethical, Business, Ecosystem, skills and technology(with Data) Each class is further detailed in a multi-level taxonomy similar to D-BEST. (Razzetti et al., 2022a)

Legal advice on AI (such as adherence to the AI Act and data protection), risk auditing and ethical evaluation for AI solutions, services guaranteeing algorithmic fairness or transparency in AI, and the creation of governance frameworks for reliable AI are all included in the Legal & Ethical class.(Razzetti et al., 2022a)

Skills include AI training programs and machine learning upskilling; Ecosystem may entail creating networks of AI solution suppliers and customers; and Technology/Data includes data platforms, test-before-invest facilities, and specialized knowledge for AI solutions.(Razzetti et al., 2022a)

Skills include AI training programs and machine learning upskilling; Ecosystem may entail creating networks of AI solution suppliers and customers; and Technology/Data includes data platforms, test-before-invest facilities, and specialized knowledge for AI solutions. By combining data and technology, L-BEST recognizes the importance of data handling in AI tech services and makes room for the new Legal-Ethical dimension(Razzetti et al., 2022a).

A hub can use the framework to benchmark a "AI DIH" by comparing its L-BEST service coverage to that of its peers, determining whether it lacks any legal or ethical advisory capabilities, and making plans to close any gaps(Razzetti et al., 2022a).

1.3.4 Alternative Frameworks for EDIH Assessment

1.4 Cluster Analysis Concepts, Methods, and Applications in EDIH Research

Cluster analysis (or *clustering*) is an unsupervised learning approach that groups a set of objects such that those in the same group (cluster) are more similar to each other than to those in other groups. (Wierzchoń & Kłopotek, 2018a). In the context of European Digital Innovation Hubs (EDIHs), clustering can reveal patterns among hubs based on their characteristics (sectors served, services offered, technologies, etc.), helping researchers and policymakers categorize hubs and identify gaps or archetypes in the EDIH ecosystem. This section provides an overview of major clustering concepts and methods – partitional, hierarchical, density-based, and model-based – and examines their applicability to EDIH data (e.g. sectors, services, technologies). It also reviews how clustering has been applied in academic and policy research on EDIHs, discusses common tools (with an emphasis on Python libraries), and evaluates which methods have proven most insightful for EDIH analysis.

1.4.1 Clustering Concepts and Methods

Partitional Clustering: Partitional methods directly partition the dataset into a pre-specified number of clusters. The classic example is *k*-means clustering, which assigns each data point to one of *k* clusters and iteratively updates cluster centroids to minimize within-cluster variances. *k*-Means aims to find clusters of roughly equal variance (often tending toward spherical clusters) by minimizing the inertia (sum of squared distances of points to their cluster centroid). It is relatively

efficient and scales well to large datasets, but it requires choosing the number of clusters *a priori* and assumes clusters are convex and isotropic. Despite these assumptions, *k*-means has been widely used across many fields due to its simplicity and speeds. (Steinley, 2006)

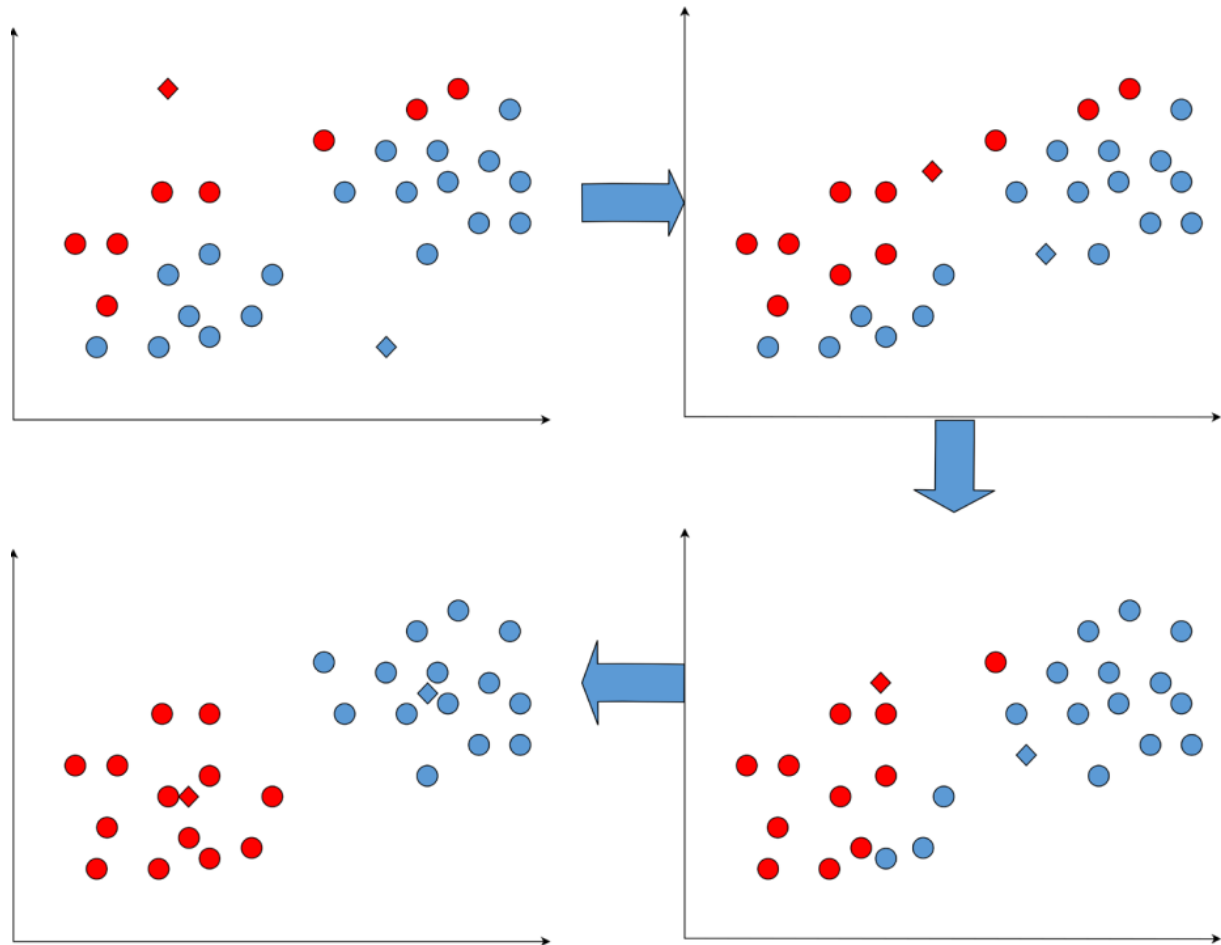


Figure 1 Schematic diagram of partition-based clustering process (Temirbekova et al., 2024)

Figure 7 visually illustrates the iterative process of partitional clustering, exemplified by the K-means algorithm. The diagram typically begins by showing an initial, often random, placement of cluster centroids within the dataset. In subsequent steps, the figure demonstrates the assignment phase, where each data point is allocated to the nearest centroid, effectively forming preliminary clusters. Following this, the figure displays the centroid update step, where the position of each cluster's centroid is recalculated based on the mean location of its newly assigned data points. This cycle of assignment and recalculation is depicted as repeating, showing the centroids gradually shifting their positions. The process continues until the cluster assignments stabilize and the centroids no longer move significantly, indicating convergence and

the final partitioned clusters. This visual progression highlights how the algorithm iteratively refines the cluster boundaries to minimize the overall within-cluster variance. (Temirbekova et al., 2024)

Hierarchical Clustering: Hierarchical clustering builds a nested hierarchy of clusters by either successively merging smaller clusters (agglomerative approach) or splitting larger clusters (divisive approach) (Wierzchoń & Kłopotek, 2018b). Agglomerative hierarchical clustering starts with each data point as its own cluster and merges clusters step by step based on a linkage criterion e.g. Ward's method minimizes within-cluster variance at each merge, analogous to k -means's objectives. The result is a dendrogram (tree) that shows clusters at various levels of granularity. This allows one to choose an appropriate number of clusters by “cutting” the dendrogram at a given level. Hierarchical methods do not require a pre-set number of clusters and can use various distance metrics (Euclidean, Manhattan, cosine, etc.) and linkage strategies. They are particularly useful for exploratory analysis, as the dendrogram can reveal the multi-level structure in the data. However, hierarchical clustering can be computationally intensive for large datasets, and agglomerative methods are *greedy* – once a merge is made it cannot be undone, so early erroneous merges can affect the final cluster structure (Wierzchoń & Kłopotek, 2018b).

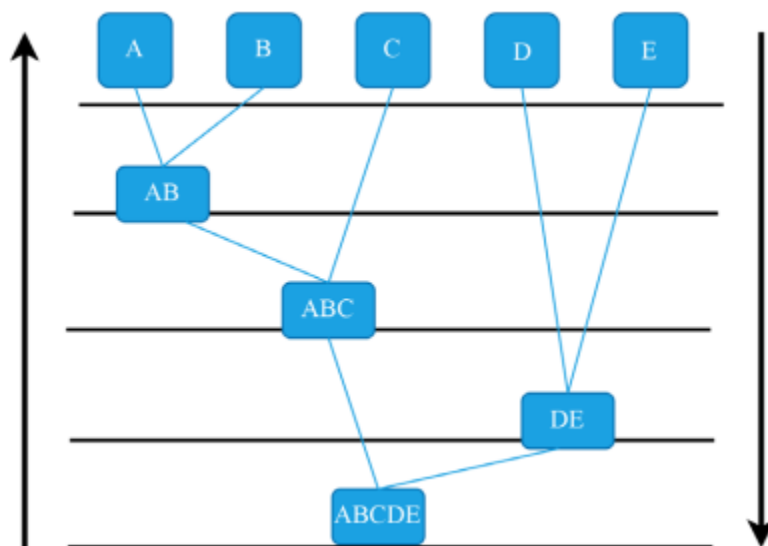


Figure 2 Schematic diagram of the hierarchical (Temirbekova et al., 2024)

Figure 8 visually represents the process of hierarchical clustering, specifically illustrating a bottom-up, agglomerative approach. It begins at the lowest level with individual data points,

labeled A, B, C, D, and E, each forming its own distinct cluster. The diagram then shows a progressive merging of these individual points and subsequent clusters based on their similarity. For instance, points A and B are first merged to form cluster 'AB'. Simultaneously, points D and E merge to form cluster 'DE'. In the next stage, cluster 'AB' merges with point C to form 'ABC'. Finally, all remaining clusters ('ABC' and 'DE') merge to form a single, overarching cluster 'ABCDE'. The vertical arrangement, with individual points at the bottom and the complete merged cluster at the top, signifies the "layer-by-layer" aggregation of data samples, culminating in a single comprehensive cluster. This structure highlights how hierarchical relationships are built by progressively combining the most similar entities.(Temirbekova et al., 2024)

Density-Based Clustering (e.g., *DBSCAN*): Density-based methods form clusters by identifying high-density regions in the data and expanding clusters from those “core” regions (Kriegel et al., 2011). *DBSCAN* (Density-Based Spatial Clustering of Applications with Noise) is a common algorithm that groups point that are closely packed (within a distance ϵ with at least *MinPts* neighbors) and marks points in low-density regions as outliers (noise) (Ester et al., 1996a). A key advantage of *DBSCAN* is that it can discover clusters of arbitrary shape – for example, clusters that are elongated or irregularly shaped, which partitional methods might struggle with. It also does not require specifying the number of clusters in advance; instead, the ϵ (neighborhood radius) and *MinPts* parameters determine cluster formation. *DBSCAN* is robust to outliers in that it naturally excludes noise points from clusters. However, it assumes a relatively uniform density for clusters – if the dataset has clusters with very different densities, a single ϵ value may not fit all, causing either over-merging or over-splitting of clusters(Ester et al., 1996a). In summary, density-based clustering is effective when clusters are irregularly shaped or when distinguishing noise is important, but parameter tuning is needed and it may struggle if density varies greatly across the data.

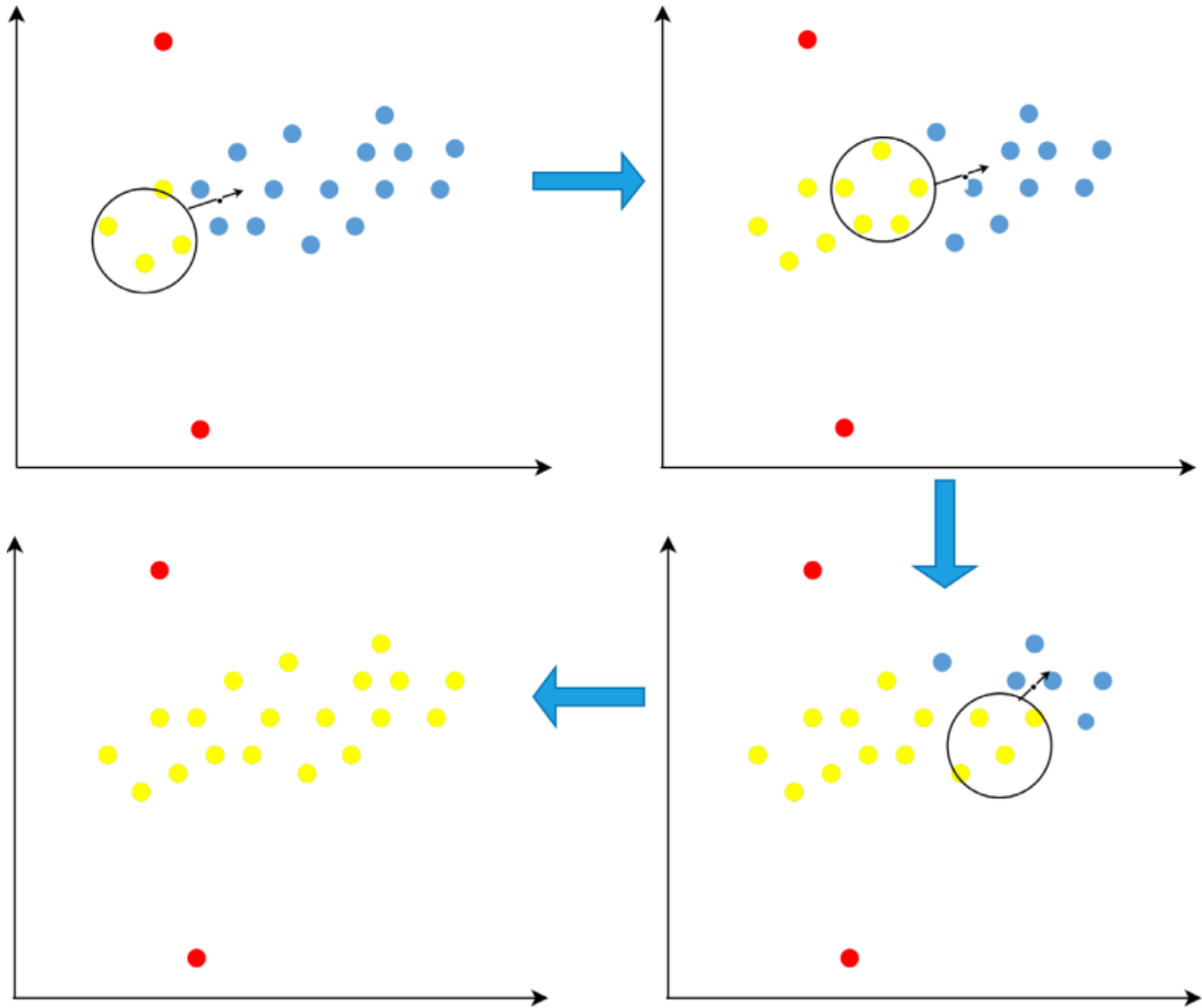


Figure 3 Diagram of DBSCAN clustering process

Fig. 5: Diagram of the DBSCAN clustering process illustrates how the DBSCAN algorithm identifies, and forms clusters based on data density. The figure shows data points, with yellow and blue points representing potential clusters and red points indicating noise. The process begins in the **top-left panel** with the arbitrary selection of a yellow data point and the establishment of its ϵ (epsilon) neighborhood. Transitioning to the **top-right panel**, if this neighborhood meets the MinPts density threshold, the central point becomes a "core point," initiating cluster growth. The algorithm then iteratively expands the cluster by incorporating all "density-reachable" points, including other core points (leading to cluster merging) and "border points" (non-core points within a core point's neighborhood), as shown in the **bottom-right panel**.

The **bottom-left panel** displays the final result: all yellow points form a single, arbitrarily shaped cluster, while the isolated red points are successfully identified as noise. This figure effectively demonstrates DBSCAN's ability to discover complex-shaped clusters and its robustness in handling outliers, by defining clusters based on local density rather than requiring a predefined number of clusters (Temirbekova et al., 2024)

Model-Based Clustering (e.g., Gaussian Mixture Models): Model-based approaches assume that the data are generated from an underlying probabilistic model – typically a mixture of probability distributions – and aim to recover those distributions. A common model-based method is the Gaussian Mixture Model (GMM), which assumes data points are generated by a mixture of a finite number of Gaussian (normal) distributions with unknown parameters (M.Bishop, 2006). Each Gaussian component represents a cluster. The model is usually fit using the Expectation-Maximization (EM) algorithm to estimate the parameters (means, covariances and weights of each Gaussian) that maximize the likelihood of the data (M.Bishop, 2006). GMM clustering is essentially a form of “soft” clustering: instead of a hard assignment of each point to a single cluster, GMM provides a probability (or membership score) of each point belonging to each cluster (Reynolds, 2009). This can capture overlap between clusters and express uncertainty in assignments. Model-based clustering can also accommodate clusters with different shapes, since each Gaussian has its own covariance (allowing elliptical clusters of varying size/orientation, unlike k -means which corresponds to Gaussians with equal spherical covariances). The trade-off is that one must either choose or infer the number of components (clusters) – methods like Bayesian Information Criterion (BIC) can help select the number of Gaussian components by trading off model fit and complexity. GMMs can be very powerful for data that indeed follow mixture distributions, but they may perform poorly or produce ill-defined solutions if the Gaussian assumption is severely violated or if there is insufficient data for some components (Reynolds, 2009). In practice, GMM clustering is slower than k -means for large datasets (due to the iterative EM and covariance calculations) and can get stuck in local optima, so it often requires multiple runs or careful initialization (M.Bishop, 2006).

1.4.2 Comparison of Clustering Methods for EDIH Data

Clustering methods have different strengths, and their suitability for EDIH research depends on the nature of the EDIH dataset. EDIHs are typically described by a mix of categorical and numeric features – for example, categorical indicators of which **sectors** they serve (manufacturing, health, agriculture, etc.), which **technologies** they focus on (AI, cybersecurity, etc.), and which **services** they offer (test-before-invest facilities, training, funding advice, etc.), as well as numeric attributes like the size of the hub or number of partner organizations (Gavkalova et al., 2024). Such data tend to be high-dimensional and often binary or ordinal (presence/absence of a sector or technology). Key considerations for clustering EDIHs include how to measure similarity on these mixed attributes, how to handle hubs that are very unique, and how to determine a meaningful grouping that can inform policy or management decisions.

Distance Measures and Data Encoding: Partitional and hierarchical clustering usually rely on a distance or similarity measure (Xu & Wunsch, 2010). For EDIH categorical data, one common approach is to encode categories as binary variables (one-hot encoding) and use a distance metric like Euclidean or Manhattan on the resulting vectors. In this case, *k*-means can be applied, but one should be cautious: Euclidean distance on high-dimensional binary data may treat hubs with different combinations of sectors/technologies as far apart even if they share some similarities. Hierarchical clustering offers more flexibility by allowing non-Euclidean distance measures – for instance, Jaccard or Hamming distance could be used to compare two hubs based on sets of sectors or services. Using an appropriate distance can improve clustering quality for categorical-heavy data. In practice, hierarchical clustering has an edge in handling mixed data since one can define a custom distance (or even use domain knowledge to weight certain features more) and then use agglomerative clustering with that distance. *k*-Means, by contrast, implicitly uses Euclidean distance in feature space (though variants like *k*-modes or *k*-prototypes exist for categorical data, they are less common). Thus, for EDIH data with many categorical fields, hierarchical clustering (or converting categories into binary and using GMM with Bernoulli mixtures) might capture structure better than vanilla *k*-means.(Reddy, 2014a)

Cluster Shape and Distribution: If EDIHs naturally group into clusters of very different sizes or non-globular shapes in feature space, partitional methods like *k*-means might struggle. *k*-Means seeks clusters that are relatively spherical and of similar radius; this might not hold if, say, one

subset of hubs shares a very broad portfolio (occupying a wide region in feature space) while another subset is tightly focused (narrow region). Hierarchical clustering with complete or average linkage can sometimes handle non-spherical clusters better (single linkage even finds chain-like clusters, though it can be sensitive to noise) (Chen, 2025). Density-based clustering (DBSCAN) could be valuable if the data contains a few dense groups of very similar hubs and some isolated hubs (Ester et al., 1996b). For example, suppose most EDIHs can be grouped into a few “types” (dense clusters in the attribute space), but a handful of hubs have unusual combinations of sector or technology focus; DBSCAN would be able to label those unique hubs as noise or separate micro-clusters, whereas k -means or hierarchical will still force them into some cluster. This outlier-detection aspect of DBSCAN is useful for finding *atypical EDIHs* (perhaps highly specialized hubs that don’t fit the mainstream patterns). On the other hand, if the EDIH data do not exhibit clear density separations (which is possible if the features are many and binary, making the data sparse), DBSCAN might either put almost everything in one cluster or everything as noise unless parameters are finely tuned. In the current EDIH network, the dataset is not huge, so identifying density-based clusters is feasible, but choosing the right ϵ distance threshold would require experimentation or domain insight (e.g. what constitutes “closely similar” two hubs in terms of sector/service profile) (Kriegel et al., 2011b).

Number of Clusters and Interpretability: A practical consideration is how to decide the number of clusters and how interpretable the clusters are. For EDIH analysis, interpretability is crucial – clusters might be used to label hubs as, say, “manufacturing-focused generalists” or “AI specialist hubs” and so on. Hierarchical clustering offers a dendrogram that analysts can visually inspect to decide a sensible cut, possibly aligning with known categories or policy needs (e.g. maybe one wants to distinguish a few broad groups of hubs)(Reddy, 2014b). k -Means and GMM require specifying k (number of clusters) in advance; one would typically run them for various k and use validation metrics like the **Elbow method** or **Silhouette coefficient** to choose an optimal k (Tibshirani et al., 2001). These techniques measure how well-separated the clusters are – for example, the Silhouette score near +1 indicates well-separated clusters. In an academic EDIH study, it’s common to experiment with different cluster counts and examine which yields meaningful groupings (perhaps cross-referencing with qualitative knowledge of the hubs) (Reddy, 2014b). In fact, a combination of hierarchical and partitional methods is often effective: hierarchical clustering (especially using Ward’s method) can be used to suggest an appropriate number of clusters by looking at the dendrogram or distance jumps, and then k -means or GMM

can refine cluster assignments using that number (Reddy, 2014b). This hybrid approach was employed by (Gavkalova et al., 2024) in clustering DIH data: they used Ward’s hierarchical method to determine that four clusters of countries was a reasonable choice, then applied k -means to finalize the clusters and characterize them.

Computational Considerations: The scale of EDIH datasets (hundreds of hubs, tens of features) is relatively small in machine learning terms. k -Means will run very quickly on this scale, and even hierarchical clustering is tractable (agglomerative clustering has worst-case complexity $O(n^3)$, but for $n \sim 500$ it is fine) (Reddy, 2014b). GMMs can also be run without performance issues on hundreds of points (M.Bishop, 2006). Thus, method choice is driven more by data characteristics and analytical goals than by computational limits. If one were to include a much larger dataset (for example, many small firms associated with EDIHs, or time-series data of EDIH activities), then scalability might favor k -means or mini-batch k -means (Reddy, 2014b). For mapping EDIHs across Europe, however, all the discussed methods are feasible. It’s worth noting that hierarchical clustering yields a lot of information (the full tree) which can be overkill; if the goal is simply to categorize hubs into a few groups, k -means or GMM with the chosen k might suffice. Conversely, if one is uncertain about the structure, exploring the dendrogram from hierarchical clustering or running a density-based scan to see if any obvious outliers exist can provide insight before settling on a final clustering scheme (Reddy, 2014b).

1.4.3 Use of Clustering in EDIH Research and Policy

Categorizing and Benchmarking EDIHs: Clustering has been employed in both academic studies and policy analyses to categorize EDIHs or related entities (like Digital Innovation Hubs, DIHs) and to map their diversity across Europe. A prominent example is a recent Joint Research Centre (JRC) technical report by the European Commission that profiled the EDIH network. While the JRC report did not explicitly run a clustering algorithm, it effectively classified hubs into types. The analysis introduced a matrix distinguishing “**Focused**” vs “**Diversified**” sector coverage and “**Specialist**” vs “**Generalist**” technology coverage, and positioned each EDIH in one of four quadrants (e.g. *Diversified Generalists* – hubs offering a wide range of technologies and serving many sectors, vs *Focused Specialists* – hubs with narrow tech focus and serving few sectors) (NIGRIS Sarah et al., 2023). Interestingly, the JRC found that larger hubs (those with many partner organizations) tended to fall in the *Diversified Generalist* category – they serve many sectors with

many technologies – whereas small hubs were more scattered, with some very specialized ones. This kind of typology is essentially a clustering of hubs along meaningful dimensions, and it provides policymakers with insight into how hubs differ. For instance, identifying a cluster of “focused specialist” hubs might indicate regions where hubs have a narrow focus – possibly reflecting regional specialization or limited capacity – which could flag a need for policy support to broaden their services(NIGRIS Sarah et al., 2023).

Academic Research Applications: In academic research on DIHs/EDIHs, formal cluster analysis has been used to draw higher-level conclusions about the digital innovation ecosystem (Gavkalova et al., 2024) performed a cluster analysis on **national-level** data about DIHs to group European economies into similar sets. They collected data on 38 countries, including the number of hubs in each country and the distribution of hub services within each country, and applied Ward’s hierarchical clustering (Euclidean distance) to identify clusters of countries. This analysis yielded four distinct clusters of economies in terms of their DIH landscape. These clusters reflected different “digitalization statuses and strategies” among the countries. For example, one cluster contained the largest Western European economies with many DIHs and a broad spread of advanced services, whereas another cluster contained smaller or less digitally developed countries with fewer hubs or more niche service profiles. Such results allow benchmarking: policymakers can compare a country to others in its cluster or to those in a more advanced cluster, to assess the relative progress of the EDIH network. The same study also noted that the clusters align with differences in regional strategies – e.g. countries in one cluster might prioritize certain services (like advanced training and testbeds) while others focus on basic awareness and networking, indicating varying roles of hubs in the innovation ecosystem. By clustering on service portfolios, the authors were able to conjecture links between cluster membership and factors like economic development or innovation policy focus (Gavkalova et al., 2024).

In summary, both policy-oriented research and academic studies have embraced clustering to make sense of the EDIH network. The **EDIH network data** (from the European Commission’s catalogue) provides the raw material – a comprehensive list of hubs with attributes – and clustering methods are applied to derive higher-order groupings. This has yielded insights such as the existence of distinct types of hubs (generalist vs specialist), the imbalance of service portfolios between different regions, and the relationship between a hub’s size or host country’s economy and the hub’s breadth of activities (Gavkalova et al., 2024). These insights are valuable for both

practitioners (to identify peer groups of hubs or potential partnerships) and policymakers (to tailor support programs according to hub cluster profiles).

1.4.4 Tools and Libraries for Clustering EDIH Data

A variety of software tools are available to perform clustering, and EDIH researchers typically leverage popular data analysis platforms. **Python** has become a go-to choice in engineering and data science research due to its rich ecosystem of libraries:

- **Scikit-learn:** The scikit-learn library is a widely used Python toolkit that implements many clustering algorithms out-of-the-box. It provides efficient implementations of k -means, agglomerative hierarchical clustering, DBSCAN, Gaussian mixture models, and more. For example, using scikit-learn, one can perform a k -means clustering in a few lines of code or generate a dendrogram from hierarchical clustering. Scikit-learn’s clustering API also supports cluster evaluation tools (e.g., functions to compute silhouette scores) and allows using custom distance matrices for hierarchical clustering. It is well-documented and optimized, making it suitable for both quick experimentation and integration into analysis pipelines (Pedregosa FABIANPEDREGOSA et al., 2011a).
- **SciPy:** SciPy, the scientific computing library in Python, complements scikit-learn by providing some lower-level clustering functions and utilities. Notably, SciPy’s hierarchy module contains linkage algorithms and dendrogram plotting functions for hierarchical clustering. Researchers can use SciPy to compute a linkage matrix (using methods like Ward, complete, average linkage) and then visualize the dendrogram to decide on cluster cuts. SciPy is also often used to precompute distance matrices (with support for various distance metrics), which can then be fed into clustering algorithms. In summary, SciPy offers the foundation (e.g., distance computations, hierarchical merges) on top of which higher-level libraries like scikit-learn build – and in some cases, SciPy’s implementations are directly used by scikit-learn under the hood (Virtanen et al., 2020).
- **PyClustering:** PyClustering is an open-source library dedicated to clustering algorithms, including several advanced or specialized methods beyond the basics. Written in C++ with Python wrappers, PyClustering provides a “wide range of clustering algorithms and methods” – for example, it includes algorithms like **OPTICS** (an extended density-based method related to DBSCAN), **X-Means** (an extension of k -means that can decide the

number of clusters), **CLARANS**, **AGNES**, and even network-based clustering algorithms. In EDIH research, PyClustering could be useful for experimenting with algorithms not in scikit-learn or for performance needs. Its focus on clustering analysis makes it a flexible tool for researchers who need to customize or dig into algorithm variants. However, scikit-learn covers most standard needs; PyClustering is typically used when a specific niche algorithm or additional performance from C++ is required (Novikov, 2019).

- **Statistical Software:** Outside of Python, researchers have used tools like **Statistica**, **R**, or **MATLAB** for clustering. For instance, (Gavkalova et al., 2024) report using Statistica 10.0 to perform Ward’s hierarchical clustering and k -means on the DIH dataset. Statistica (a commercial analytics software) and IBM SPSS provide user-friendly interfaces for clustering, including visual outputs like dendrograms and silhouette plots, and can automatically apply methods like the elbow criterion to suggest cluster numbers. R, with packages like *cluster* or *facto extra*, is also heavily used in academic research to do clustering analysis and visualization (dendrograms, clustering validation statistics, etc.). The choice of tool often comes down to user preference and the broader analysis workflow; in modern EDIH studies, Python is common for its ability to integrate data preprocessing (e.g. web scraping of hub info, text mining of descriptions) with clustering in one environment.

Regardless of the platform, certain **practices** are crucial. Data preprocessing (cleaning, standardizing scales, handling missing values) is typically done before clustering. For example, if one feature is “number of services offered” (a count) and another is a binary flag for “offers AI-related services or not,” these have different scales; one may choose to standardize numeric features or weight features to avoid one type dominating the distance calculations. Feature selection or dimensionality reduction (like using PCA) can also be applied if there are very many features (though in EDIH analysis the feature count is moderate). Finally, validation of clusters is important: researchers will often compute internal validation metrics (e.g., silhouette scores) or examine cluster cohesion/separation, and also **interpret clusters with external information** – for example, checking if clusters correlate with geographical regions or known classifications. The integration of qualitative domain knowledge with clustering results is a common theme in EDIH research, ensuring that the clusters “make sense” in real-world terms (e.g., do the hubs in Cluster indeed share a common theme that an expert recognizes?) (Reddy, 2014b). Tools like scikit-learn

and R facilitate this by allowing easy extraction of cluster members for further analysis and plotting (Pedregosa FABIANPEDREGOSA et al., 2011b).

1.4.5 Effectiveness of Clustering Methods for EDIH Analysis

Different clustering methods offer different insights in the EDIH context, and researchers have reported varying success with them. Overall, a combination of hierarchical and partitional clustering has emerged as an effective strategy for many EDIH analyses. Hierarchical clustering (especially using Ward's linkage) is often used in an exploratory phase because it can reveal the possible number of clusters and the relationships between individual hubs or countries (Gavkalova et al., 2024). Ward's method in particular has been praised for producing balanced, interpretable clusters in socio-economic data, as it minimizes variance within clusters (Comaniciu & Meer, 2002); (Gavkalova et al., 2024) note that Ward's hierarchical clustering gave them a clear initial dendrogram which suggested four meaningful clusters of economies, which they then solidified with k -means. This indicates that **hierarchical clustering is very insightful** when one does not know the structure beforehand – it provides a full picture from 1 cluster (all hubs together) to N clusters (each hub separate), and one can choose the level that yields a sensible grouping. The downside – that hierarchical methods can be sensitive to noise – has not been a major issue in EDIH studies so far, because the datasets (e.g., national aggregates or curated hub profiles) are not extremely noisy and are of manageable size (Reddy, 2014b).

k -Means remains a **popular, effective method** for final clustering once the desired number of clusters is chosen. It's easy to interpret (each cluster can be described by its centroid – e.g., “Cluster A's centroid has high values on manufacturing and IoT features, indicating those hubs emphasize those areas”). k -Means also allows easy assignment of new observations (if a new hub comes in, one can quickly see which centroid it's closest to). In EDIH research, where clarity and simplicity are valuable for communicating to policymakers, k -means has been useful. However, one must ensure its assumptions are not violated; if an EDIH cluster is expected to be non-spherical or very different in size, k -means may lump or split clusters in ways that don't match reality (Comaniciu & Meer, 2002). So far, studies that used k -means (either alone or after hierarchical) reported sensible results, likely because they had identified reasonably well-separated clusters to begin with (e.g., the four country clusters in the 2024 study were sufficiently distinct in the chosen features to be captured by k -means) (Gavkalova et al., 2024). In cases where k -means was applied to hubs

directly, researchers often reduce dimensionality or cluster on summary indices to satisfy the method's requirements.

Density-based methods like DBSCAN have been less frequently reported in published EDIH analyses, but they hold potential. The relative rarity might be because the number of EDIHs is not extremely large and analysts can manage outliers in other ways (or simply note them qualitatively) (Kriegel et al., 2011a). Moreover, DBSCAN's results (number of clusters discovered) can be unstable if the distance threshold is not set well, and explaining a DBSCAN cluster to stakeholders is less straightforward than saying "these are cluster 1 vs cluster 2 from *k*-means." That said, DBSCAN could be insightful for *discovering if any hubs are outliers* (Bushra & Yi, 2021). If the EDIH data includes geographical coordinates or other spatial attributes, DBSCAN can naturally cluster by region if needed (though geographic clustering of hubs is usually obvious from country groupings). In summary, while not yet prominent in literature, density-based clustering is a method that could complement *k*-means/hierarchical by highlighting the density structure of the data – future analyses with larger numbers of hubs or more granular data might make use of it. (Ester et al., 1996b)

Model-based clustering (GMMs) has so far been relatively uncommon in documented EDIH research. This may be due to the complexity it adds – the need to estimate probabilistic models might be beyond the immediate needs of policy studies that favor straightforward segmentation. Additionally, many features of EDIHs are categorical, which GMMs (in their standard form assuming Gaussian components on continuous data) do not natively handle. One could use GMMs on numeric indicators (e.g., a continuous index of "diversity of services" and "number of technologies offered"), which might reveal latent group structure with overlapping clusters. If, for instance, the boundaries between hub types are fuzzy (as one might expect – an EDIH could reasonably belong partly in two categories), GMM soft assignments would acknowledge that. This could provide a more nuanced view than hard clustering. However, interpreting and communicating probabilistic clusters is more challenging. So far, studies have found that hard clustering into clear groups (even if somewhat artificial boundaries) is more actionable for policy purposes. We anticipate that as the EDIH network grows and data become richer, model-based approaches might be explored (especially with techniques to handle mixed data, or using Dirichlet processes to let the number of clusters be inferred). But at present, their absence in literature

suggests that simpler methods already suffice for the main insights needed (NIGRIS Sarah et al., 2023).

In evaluating which method is **most effective**, it truly depends on the analytical goal. For *exploratory mapping* of the EDIH landscape, hierarchical clustering has proved invaluable, since it does not force a single grouping and can reveal hierarchical relationships (e.g., one might see that all hubs in a certain country first cluster together before merging with others, indicating country-specific traits – something *k*-means wouldn't show without multiple runs)(Reddy, 2014b). For *concise description and benchmarking*, a fixed partition (using *k*-means or an equivalent) with a sensible *k* is effective and easier to communicate (e.g., in a report: “We identify three clusters of hubs across Europe...”). If *novel patterns or anomalies* are of interest, density-based methods add an additional layer of insight by not assuming every point must belong to a cluster. Ultimately, many EDIH studies have adopted a **multi-method approach**: using hierarchical clustering to guide the choice of clusters, *k*-means to assign clusters for characterization, and then validating or interpreting those clusters with domain knowledge. This approach has been effective in producing robust and meaningful cluster-driven analyses of the EDIH ecosystem (Gavkalova et al., 2024).

2 Methodology

This chapter outlines the systematic approach undertaken to analyze the European Digital Innovation Hub (EDIH) ecosystem, culminating in the development of a novel Sector-Service-Technology (SST) matrix. The methodology is designed to ensure comprehensive preprocessing, and the application of advanced analytical techniques to uncover nuanced understandings of EDIH specializations and operational patterns across Europe. The research adopts a comprehensive, multi-methodological approach to systematically understand the regional. Distribution, sectoral specialization, technological focus, and functional diversity of EDIHs. The overarching objective is to uncover analytical understandings into EDIHs by analyzing their service offerings using the SST matrix, which provides a novel visualization of each EDIH's offerings across sectors, services, and technology domains.¹ This design is structured around three core objectives: comprehensive EDIH mapping, development of the SST matrix, and identification of EDIH typologies.¹ The overall research flow, as depicted in the methodology diagram, progresses from conceptualization and systematic literature review through EDIH analysis, data collection, preprocessing, keyword analysis, geo-spatial map visualization, framework development, and culminates in cluster analysis and policy recommendation generation.

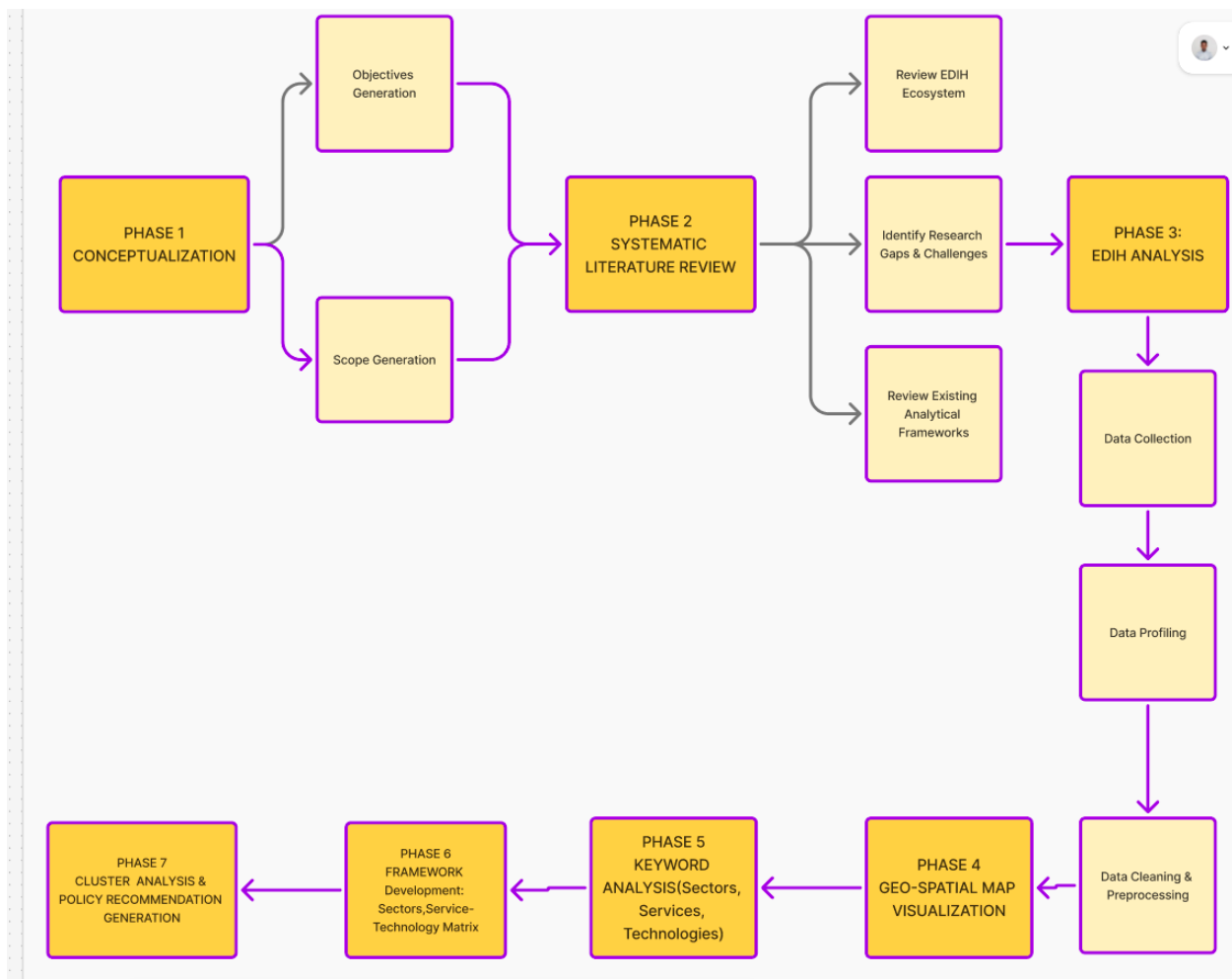


Figure 4 Research Methodology Diagram

2.1 Data Collection

The foundation of this research relies on a multi-tiered data collection strategy, integrating primary and secondary sources to construct a robust and comprehensive dataset for analysis. The primary data source was the official European Digital Innovation Hub (EDIH) Catalogue, published by the European Commission and accessible through its digital portal. The catalogue was downloaded in both Excel and CSV formats (export-edihs.xls) to allow flexibility for preprocessing and analysis workflows. It contains detailed records of Digital Innovation Hubs (DIHs) operating across Europe, capturing critical attributes such as hub name, location, coordinating institution, country, region, sectors served, technologies applied, services offered, a free-text “Description” field outlining missions and activities, and links to hub websites.

However, recognizing that the catalogue data was occasionally incomplete or inconsistently formatted, strict eligibility thresholds were defined. Only EDIHs with non-missing, structured data for key fields—specifically sectors served, services offered, and technologies applied—were qualified for further analysis. This selective approach aimed to ensure that subsequent analytical steps would rely on robust and comparable data across hubs.

In parallel, secondary datasets were consulted to enrich the hub-level data with macro-level indicators. These included national and regional statistics from Eurostat, digital transformation scores from the Digital Economy and Society Index (DESI) reports, and relevant policy frameworks published by the European Union.

Finally, qualitative inputs were collected from whitepapers, national digital strategies, and innovation roadmaps to contextualize the dataset and facilitate a later triangulation of quantitative and qualitative findings. This multi-tiered data collection strategy allowed the construction of a nuanced, comprehensive database suited for both statistical and content-based analysis.

2.2 Data Preprocessing

Upon completing the initial data collection phase, a rigorous data preprocessing workflow was undertaken to prepare the dataset for structured analysis. Data preprocessing is a critical step in ensuring the quality, consistency, and analytical tractability of the dataset.

It is important to note that the source dataset was obtained as a spreadsheet (Excel format), with the most recent download occurring on May 5, 2025. The distribution of EDIHs before the next step of the methodology is plotted in table.

Defining Eligibility Conditions for EDIH Inclusion

To ensure that subsequent analytical steps would rely on robust and comparable data across hubs, strict eligibility thresholds were defined and applied in a multi-stage filtering process. This selective approach directly addresses limitations of previous research, such as reliance on potentially biased self-reported data and limited granularity in analysis. The sequential application of these conditions allowed for a systematic refinement of the dataset.

Condition 1: Core Data Availability for Initial Filtering

Definition: Each EDIH record was required to have data available in four specific columns: Description (indexed field), Formatted Sectors, Formatted Services, and Formatted Technologies. Any EDIH records lacking information in any of these fields were flagged for exclusion or further investigation.

Purpose: This initial condition served as a fundamental completeness check. It identified EDIHs that either had sufficient data for immediate consideration or required further data collection (e.g., through web scraping) to become potentially eligible.

Impact: After this initial assessment, the dataset's coverage based on these core categories was 62.8% of the initial entries. This indicated a substantial portion of the initial raw data was incomplete, necessitating further data acquisition efforts, particularly web scraping, to fill critical gaps.

Condition 2: Enhanced Data Completeness and Accessibility

Definition: To qualify for further analysis, an EDIH needed to meet a less stringent data completeness requirement for its core categorical fields, specifically having data available for at least two of the following categories: Formatted Sectors, Formatted Services, or Formatted Technologies. Crucially, this condition also mandated the presence of a Valid Website for the EDIH.

Purpose: This condition aimed to ensure that EDIHs included in the study had a verifiable online presence and a minimum level of descriptive information regarding their core offerings. The requirement for a valid website was particularly important for validating self-reported data and for future data enrichment, allowing for more reliable data validation and subsequent analysis.

Impact: This condition expanded the eligible pool, resulting in an EDIH coverage of 64.99% and encompassing 37 EU Member States. This step marked a significant improvement in data quality and breadth, moving closer to the desired pan-European scope.

Condition 3: Final Eligibility for Research Study

Definition: This final condition refined the criteria from Condition 2, reiterating the requirement for at least one of the Formatted Sectors, Formatted Services, or Formatted Technologies categories to be populated, alongside a Valid Website. While the textual description appears identical to Condition 2, the slight improvement in coverage from 64.99% to 65.2% indicates a

final pass or minor adjustments in the data qualification process, possibly due to successful web scraping efforts or a final manual review of borderline cases.

Purpose: This condition established the definitive set of EDIHs deemed sufficiently complete and reliable for the deep research study. It represents the final selection after all data cleaning, scraping, and verification efforts, ensuring the analytical findings would be statistically sound and meaningful.

Impact: The application of Condition 3 resulted in a final EDIH count of 298, representing 65.2% of the initial entries, and encompassing all 38 EU Member States targeted for the study.

While these efforts initially brought the total EDIH count to 298, it was subsequently discovered that the website for one of these EDIHs was inactive, rendering web scraping impossible and leading to its automatic exclusion from the dataset due to insufficient data. Consequently, the final dataset for the study comprises 297 EDIHs, representing 64.99% of the initial 457 entries, while successfully ensuring representation from all 38 EU Member States.

The progression from an initial 457 EDIH entries to a post-cleanup count of 297, achieved through the sequential application of Condition 1, Condition 2, and Condition 3, signifies a highly rigorous and iterative data curation process. The substantial reduction in the dataset size (159 entries) is a direct consequence of these stringent eligibility criteria. This indicates that a significant portion of the initial raw data was either incomplete, inconsistent, or lacked a verifiable online presence, underscoring the importance of robust data validation. This meticulous filtering enhances the reliability and comparability of the final dataset, which is crucial for the validity of subsequent analyses, such as clustering and SST matrix development. The process also highlights a potential systemic challenge within the broader EDIH ecosystem: the variability in data reporting and digital presence among hubs, which can impede comprehensive analytical efforts.

Table 7, depicts the distribution of EDIHs by country both initially and after the completion of the comprehensive data preprocessing, which involved the application of three rigorous eligibility conditions. This table provides a clear quantitative overview of the final dataset used for the research study, highlighting the impact of the data cleaning process on each nation's representation.

Albania	4	4	0.0%
Austria	11	4	63.6%
Belgium	15	7	53.3%
Bosnia and Herzegovina	2	2	0.0%
Bulgaria	13	11	15.4%
Croatia	17	5	70.6%
Cyprus	2	2	0.0%
Czechia	11	8	27.3%
Denmark	6	5	16.7%
Estonia	2	1	50.0%
Finland	9	4	55.6%
France	30	16	46.7%
Germany	57	42	26.3%
Greece	9	9	0.0%
Hungary	12	9	25.0%
Iceland	1	1	0.0%
Ireland	6	4	33.3%
Italy	57	36	36.8%
Kosovo	1	1	0.0%
Latvia	4	3	25.0%
Liechtenstein	2	2	0.0%
Lithuania	10	4	60.0%
Luxembourg	1	1	0.0%
Malta	2	2	0.0%
Montenegro	1	1	0.0%
Netherlands	16	9	43.8%
North Macedonia	4	4	0.0%
Norway	3	3	0.0%
Poland	21	14	33.3%
Portugal	17	12	29.4%
Romania	9	7	22.2%
Serbia	6	5	16.7%
Slovakia	6	5	16.7%
Slovenia	10	5	50.0%
Spain	51	27	47.1%

Sweden	12	6	50.0%
Türkiye	7	7	0.0%
Ukraine	10	10	0.0%
Total	457	298	

Table EDIH Distribution Pre and Post Data Preprocessing by Country

All the subsequent data analysis was completed using Python, leveraging the powerful **pandas** and **numpy** libraries for efficient data manipulation, transformation, and numerical operations.

Key technologies and libraries utilized include:

- **Data Manipulation and Preprocessing:** pandas for data structuring and cleaning, numpy for numerical operations.
- **Web Scraping (for data enrichment during preprocessing):** Selenium for browser automation and BeautifulSoup for parsing HTML content.
- **Natural Language Processing (NLP):** nltk for text tokenization and lemmatization, scikit-learn for TF-IDF vectorization and Latent Dirichlet Allocation (LDA) for topic modeling.
- **Statistical Analysis and Machine Learning:** scikit-learn for various clustering algorithms (e.g., K-Means, hierarchical clustering) and statistical modeling. SciPy was used for advanced scientific computing, including hierarchical clustering and distance metrics. PyClustering provided additional specialized clustering methods.
- **Data Visualization:** matplotlib and seaborn were employed for creating informative plots, heatmaps, and other visual representations of the data and analytical results.

The entire codebase and analytical scripts are publicly available on GitHub at: https://github.com/amanuelgthn/EDIH_MASTER_THESIS. This ensures transparency, reproducibility, and allows for further exploration and validation of the methods employed.

2.3 Analytical Methods

To derive actionable insights from the structured dataset, a range of analytical methods was employed, blending quantitative statistical techniques with qualitative interpretive approaches. These methods were selected to address both the complexity and the multidimensionality of the

research problem: understanding the specialization patterns, regional distributions, and strategic alignments of EDIHs across Europe.

First, Text Mining and Natural Language Processing (NLP) techniques were applied to the free-text “Description” fields associated with each hub. After text cleaning and preprocessing, the descriptions were vectorized using Term Frequency–Inverse Document Frequency (TF-IDF) transformation. Topic modeling was conducted using Latent Dirichlet Allocation (LDA) to identify clusters of commonly occurring themes and technological focuses among the hubs. Additionally, keyword co-occurrence networks and word clouds were generated to visualize thematic distributions and the interconnectedness of services and sectors.

Second, Sector and Technology Profiling was conducted. Sectoral and technological features were encoded as categorical variables, and statistical visualizations such as frequency distributions, cross-tabulations, and heatmaps were used to uncover dominant sector-technology pairings. These profiles provided a granular view of how EDIHs differentiate themselves across different industrial and technological domains.

Third, Cluster Analysis was employed to group similar EDIHs. Algorithms such as k-means clustering, hierarchical agglomerative clustering, and DBSCAN were used depending on the structure of the data. Clustering was based on combinations of sector focus, technological specialization, and service breadth, as well as embedding representations from text descriptions.

Fourth, Statistical Modeling approaches were used to establish relationships between hub characteristics and external macro-level variables. Multivariate regression models assessed the predictive power of technology and sector breadth for external indicators such as national DESI scores. Logistic regression models explored the probability of cross-border service provision among hubs.

Fifth, Qualitative Case Studies were carried out for selected EDIHs based on regional diversity, specialization focus, and maturity stage. For these hubs, detailed website content analysis and strategic document comparison were performed.

Finally, a Content Analysis of Strategic Documents was undertaken to identify the major policy drivers influencing EDIH specialization patterns. Coding and thematic analysis techniques were used to systematically evaluate national digital strategies and innovation frameworks.

Collectively, these methods ensured a robust, multifaceted exploration of the European EDIH landscape.

2.4 Framework Development

Building upon the findings of the analytical stages, two strategic frameworks were developed to systematize the understanding of specialization patterns, service capabilities, and regional strengths across the EDIHs.

The first framework developed was the SST Matrix (Sector–Service–Technology). A three-dimensional mapping of sectors, services, and technologies was constructed for each EDIH, creating a comprehensive profile of their specialization spaces. The SST Matrix allowed visualization of specialization patterns not only at the individual hub level but also across regions and countries. Through this matrix, clusters of high-service-density sectors and emerging technology fields were easily identified, aiding in the detection of both regional strengths and service delivery gaps.

The second major outcome was the refinement and expansion of the D-BEST Framework. Originally designed to benchmark DIHs, the D-BEST classification was enhanced based on insights from text mining, sector-technology profiling, and content analysis. New dimensions were proposed to better capture evolving service capabilities among hubs, including:

Legal advisory services, recognizing the increasing importance of regulatory compliance in digital transformation,

Remote delivery capabilities, especially relevant in the post-pandemic digital economy,

Ethical AI guidance, addressing the growing need for responsible AI deployment.

This expanded D-BEST Framework provides not only a richer lens for evaluating EDIHs but also serves as a tool for strategic planning, policy alignment, and future benchmarking initiatives across Europe.

Both frameworks were validated through pilot applications on subsets of the dataset and iteratively refined based on feedback from preliminary results. Thus, the frameworks produced are deeply

grounded in empirical evidence and aligned with emerging trends in digital innovation and regional policy.

To extract insights from the collected data, the following analytical approaches are applied:

Text Mining and NLP-Based Content Analysis:

Unstructured text in the "Description" column is analyzed using natural language processing (NLP) techniques. Steps include cleaning, tokenization, lemmatization, stop-word removal, and transformation into numerical vectors using TF-IDF.

Topic modeling (LDA) helps identify dominant service themes or technological trends.

Keyword co-occurrence networks and word clouds are used to visualize thematic clustering.

Sector and Technology Profiling:

Sectoral and technological attributes are encoded as categorical variables.

Frequency distributions, cross-tabulations, and heatmaps help identify dominant sector-technology pairings.

Cluster Analysis:

EDIHs are grouped using k-means, hierarchical clustering, and DBSCAN to identify clusters of similar hubs.

Clustering is based on encoded variables including sector, technology, service count, and description embeddings.

Tools used: Python's scikit-learn, SciPy, and Statistica.

Statistical Modeling and Regression Analysis:

Multivariate regression is applied to evaluate how variables such as technology focus or sector breadth predict external indicators (e.g., DESI scores).

Logistic regression may be used to examine the probability of a DIH having cross-border operations based on specialization profiles.

Qualitative Case Studies and Comparative Review:

Selected DIHs with varying levels of maturity, geography, and specialization are studied in-depth. Website analysis, and comparison with regional innovation strategies are conducted to understand contextual differences.

Content Analysis of Strategic Documents:

Policy documents are analyzed using coding and thematic analysis to identify key policy drivers. Strategic alignment between policy priorities and DIH specialization is evaluated.

3. Framework Development

SST Matrix (Sector–Service–Technology):

A three-dimensional matrix is built to map the relationships among sectors, services, and technologies for each DIH. This allows identification of specialization patterns and regional strengths or gaps.

Refinement of the D-BEST Framework:

Text mining and SST analysis support the development of an expanded D-BEST classification. New dimensions such as legal advisory, remote delivery, and ethical AI guidance are proposed. This expanded framework enhances benchmarking, strategic planning, and future evaluation efforts.

3 Results

- a. Sectoral & technological mapping of DIHs
- b. Clustered DIH classifications based on services & economic priorities
- c. Statistical modeling findings on key drivers of DIH success

4 Discussion

- d. Alignment of DIHs with industry needs
- e. Gaps in green technology adoption
- f. Impact of EU policies on DIH effectiveness
- g. Operational challenges and stakeholder engagement issues

5 Conclusion

- h. Summary of key findings
- i. Policy recommendations for improving DIH impact
- j. Recommendation for future studies

5.1.1 Research Objectives and Contributions

This research aims to address the following key objectives:

1. Sector-Specific DIH Activities: Assess how DIHs align with industry-specific digital transformation needs and whether their service offerings match sectoral demands.
2. Technology Adoption Patterns: Examine regional disparities in technology adoption, particularly in AI, IoT, blockchain, cybersecurity, and cloud computing, and how economic priorities shape these trends.
3. Policy Influence and Operational Challenges: Investigate how EU and national policies impact DIH effectiveness and explore the challenges DIHs face in stakeholder engagement, scalability, and funding sustainability.
4. Circular Economy integration

This study contributes to existing literature by:

- Expanding the D-BEST clustering model to include sectoral and technological variables.
- Developing a Sector-Service-Technology Matrix to map DIH offerings to industry needs.
- Providing policy recommendations to enhance DIH effectiveness, particularly in underperforming regions.

5.1.2 Methodology

A mixed-methods approach is used to integrate quantitative and qualitative insights. The research consists of the following key methodological steps:

1. Quantitative Data Collection & Analysis

a. Data Sources:

- i. EDIH catalogue for extracting information about EDIH
- ii. Eurostat & DESI: Provides socioeconomic indicators (GDP, digital readiness, innovation capacity).
- iii. Web Scraping: Uses text mining on DIH websites to identify sectoral and technological focus areas, current plan is to automatically scrap DIH websites and look for key words to categorize their service and technology implemented

b. Methods to be Applied:

- i. Cluster Analysis: Expands existing geographic clustering models to include sectoral and technological classifications.
- ii. Statistical Modeling: Identifies factors influencing DIH service diversity, including funding sources, sector demands, and policy alignment.

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